

Probabilistic Channel Attribution via Stacked Inference: Leveraging Temporal Dynamics and Creative Intelligence for Optimal Marketing Budget Reallocation

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Abstract—This paper develops a two-stage Bayesian framework for decoding the incremental contribution of multi-channel advertising to business outcomes in a privacy-first environment. The approach combines time-varying latent effects through a stacked architecture: an initial stage learns carryover (delayed impact) and reach (diminishing returns) patterns using MCMC-based Delayed Saturated models, followed by a kernel-based temporal regression layer that captures channel-specific effectiveness drift across time windows. Integration of LLM-derived semantic features from creative content enables richer signal extraction beyond traditional spend metrics, enriching the feature space for more nuanced attribution. The model outputs both predictive performance forecasts and interpretable channel contribution posteriors, which feed into constrained optimization routines (SLSQP) to recommend actionable budget reallocations aligned with business constraints. Experimental validation on synthetic and real-world marketing spend data demonstrates superior posterior parameter recovery compared to naive baselines, with results grounded in AB-test incrementality as external validation. The framework provides insurance and digital-first enterprises with a principled, uncertainty-aware approach to marketing ROI quantification and cross-channel budget allocation without user-level tracking.

Index Terms—Bayesian inference, marketing mix modeling, channel attribution, time-varying coefficients, budget optimization, MCMC, variational inference.

I. INTRODUCTION

A. The Attribution Challenge in Privacy-First Marketing

The digital advertising ecosystem has undergone a fundamental transformation. Regulatory frameworks including GDPR and CCPA, combined with platform-level restrictions on third-party cookies and mobile device identifiers, have systematically dismantled the user-level tracking infrastructure that powered attribution for two decades. For insurance carriers like Lemonade, which operate entirely through digital acquisition channels, this shift presents an existential methodological challenge. Traditional attribution methods—last-click, multi-touch, even algorithmic attribution—depend on stitching user journeys across devices and sessions. When identifiers vanish, the journeys fragment.

Marketing mix modeling offers an alternative path. Rather than tracking individuals, MMM aggregates spend and outcome data at temporal granularity: daily, weekly, or monthly spend by channel, paired with business outcomes such as

policies issued or lifetime value acquired. This aggregation operates entirely on anonymized, privacy-preserving data. The trade-off is methodological: we must infer causal contributions from observational time series without the benefit of randomized experiments or individual-level controls.

B. The Insurance Context: Why Lemonade Requires Specialized Treatment

Insurance marketing differs fundamentally from consumer packaged goods or e-commerce. The purchase cycle is longer, consideration is deeper, and the decision involves substantial financial commitment. A customer may view a television advertisement, conduct brand-agnostic search, read reviews, visit the website multiple times, and finally purchase after two weeks of deliberation. Each touchpoint contributes to the eventual conversion, but their effects are distributed in time and interact non-linearly.

Traditional MMM approaches, developed primarily for fast-moving consumer goods, treat advertising effects as contemporaneous or following simple geometric decay. Our analysis of Lemonade’s historical data revealed more complex patterns. Television spend exhibits carryover extending three to four weeks, with peak impact occurring not in the first week but the second. Social media display shows rapid saturation beyond modest spend thresholds. Brand search, paradoxically, responds both to own-channel spend and to upper-funnel activity—customers who see television commercials subsequently search for Lemonade by name.

C. Gaps in Existing Bayesian MMM Literature

Recent advances in Bayesian MMM, particularly Google’s Lightweight MMM and Meta’s Robyn, have democratized access to sophisticated attribution methods. PyMC-Marketing provides an accessible, opinionated implementation that has been widely adopted. However, these tools share common limitations. First, they assume channel effectiveness remains constant over time, contradicting both marketing theory and empirical observation. Creative fatigue, competitive response, and seasonal audience composition shifts all cause channel performance to drift. Second, their optimization routines produce point recommendations without uncertainty quantifica-

tion, inviting overconfident reallocation decisions. Third, they treat spend as the sole driver of channel effectiveness, ignoring the creative content itself.

D. Our Contribution: Stacked Inference with Temporal Dynamics and Creative Features

This paper presents a stacked Bayesian architecture that addresses these limitations. Our approach separates the modeling problem into two stages. Stage one learns channel-specific carryover and saturation parameters using MCMC sampling within a Delayed Saturated MMM framework. These parameters are structurally identified and transferable across time periods. Stage two transforms the raw spend data using these learned parameters, then fits a kernel-based time-varying coefficient model that captures weekly fluctuations in channel effectiveness.

The stack delivers three advantages. Carryover and saturation parameters benefit from the full posterior exploration of MCMC, including uncertainty quantification and multimodal distributions when present. The temporal coefficients, estimated via variational inference, scale efficiently to daily data and hundreds of channels. Critically, the additive structure preserves linearity in the transformed features, enabling straightforward contribution decomposition and SLSQP-based budget optimization under business constraints.

E. Creative Intelligence: Beyond Spend-Centric Attribution

We integrate creative intelligence by embedding advertisement copy and creative assets using transformer-based language models. These embeddings are projected onto a low-dimensional semantic space via PCA and incorporated as time-varying covariates interacting with channel spend. This formulation allows the model to distinguish between campaigns with different creative strategies and to learn which semantic dimensions correlate with higher incremental response.

F. Structure and Roadmap

The remainder of this paper proceeds as follows. Section 2 formalizes the problem and presents the stacked Bayesian architecture, including prior specification, likelihood, and inference strategy. Section 3 details the creative intelligence pipeline and integration with spend data. Section 4 describes experimental validation on synthetic benchmarks and real-world Lemonade data, including comparison against baseline methods and external validation via geo-based A/B tests. Section 5 presents the budget optimization formulation and empirical reallocation recommendations. Section 6 discusses limitations and outlines extensions including geo-level hierarchies and nested funnel effects. Section 7 concludes with implications for marketing measurement in the privacy-first era.

II. RELATED WORK

A. Foundations of Marketing Mix Modeling

The intellectual origins of modern marketing mix modeling trace to the pioneering work of Broadbent in the 1970s, who

first formalized the relationship between advertising spend and sales response using distributed lag models. These early specifications recognized that advertising effects are not instantaneous but persist across time, a phenomenon captured through geometric decay structures. The carryover parameter α introduced in Section II-B inherits this intellectual lineage, though our Bayesian treatment diverges from the classical econometric approach that treated parameters as fixed but unknown [1], [2].

The saturation transformation, which models diminishing returns to advertising intensity, emerged from the empirical observation that response functions are concave. Early practitioners employed logarithmic and square-root transformations; contemporary practice favors the Hill function for its interpretable half-saturation parameter. Our sigmoid formulation represents a departure from this consensus, offering improved numerical stability and simpler prior specification while preserving the essential concave shape [3]–[5].

The integration of carryover and saturation into a unified adstock function was codified in Google’s Lightweight MMM framework and subsequently adopted by PyMC-Marketing. These implementations demonstrated that compositional order—carryover preceding saturation—matters for both identifiability and predictive performance. Our stacked architecture extends this insight by separating parameter learning from temporal coefficient estimation, a distinction absent from end-to-end Bayesian implementations [6]–[8].

B. Bayesian Approaches to Attribution Uncertainty

The Bayesian paradigm entered marketing mix modeling through the work of Rossi and Allenby in the 1990s, who emphasized that managerial decisions require not only point estimates but full predictive distributions. Despite this methodological advance, commercial MMM practice remained dominated by ordinary least squares and gradient boosting machines that produce no coherent uncertainty quantification [9]–[11].

Recent open-source contributions have democratized Bayesian MMM. Meta’s Robyn implements a hybrid approach combining ridge regression with evolutionary optimization, producing pseudo-Bayesian uncertainty intervals through bootstrap aggregation. PyMC-Marketing provides a fully Bayesian specification with customizable priors and MCMC sampling. Both frameworks, however, assume time-invariant coefficients, limiting their ability to detect creative fatigue, competitive response, or seasonal audience composition shifts [12]–[14].

Our hybrid inference strategy—MCMC for structural parameters, variational inference for temporal coefficients—occupies a middle ground. Full MCMC on the complete model with 12 channels and 156 weeks exceeds 12 hours of sampling time, impractical for weekly retraining cycles. Mean-field ADVI alone cannot capture multimodal posteriors for carryover parameters when identification is weak. The stack resolves this tension, an approach validated in adjacent domains including healthcare information

exchange and financial transaction monitoring where similar computational constraints apply [15]–[17].

C. Time-Varying Coefficient Models in Marketing

The assumption of stationary channel effectiveness is increasingly untenable. Creative fatigue reduces response rates as audiences tire of identical executions. Competitive activity suppresses marginal returns through bidding pressure. Seasonal shifts in audience composition alter the demographic profile of reached users. Despite this, commercial MMM implementations overwhelmingly assume constant coefficients [18]–[20].

The marketing literature offers alternatives. Time-varying parameter models with Kalman filtering have been proposed but require strong parametric assumptions about coefficient evolution. Regime-switching models detect discrete shifts but cannot capture gradual drift. Our kernel-based temporal regression, adapted from the KTR framework developed for forecasting applications, imposes smoothness through radial basis functions while permitting flexible, data-driven trajectories. This representation scales efficiently to dozens of channels and hundreds of weeks [21]–[23].

The connection between time-varying coefficients and budget optimization is underexplored. If channel effectiveness declines over time, optimal allocation should front-load spend before effectiveness erodes. Conversely, if effectiveness increases due to learning effects, delayed investment maximizes return. Our optimization routine, which propagates full posterior distributions of $\beta_{t,p}$ through SLSQP, quantifies the decision risk associated with temporal uncertainty—a contribution absent from both academic literature and commercial practice [1]–[3].

D. Creative Analytics and Semantic Feature Integration

Marketing spend is an impoverished descriptor of advertising activity. Two advertisements with identical budgets, placements, and target audiences generate substantially different response rates based on creative execution. Despite this, spend remains the exclusive input to virtually all attribution models; creative quality is treated as unobserved error [4]–[6].

The computational advertising community has developed sophisticated methods for creative optimization through online A/B testing. These approaches optimize within-channel execution but do not inform cross-channel allocation. Conversely, MMM informs allocation but ignores creative variation. Our work bridges this gap by embedding creative assets into the attribution model itself [7]–[9].

Transformer-based language models provide a mechanism to extract semantic features from ad copy at scale. BART, RoBERTa, and other encoder architectures produce dense vector representations that capture message framing, emotional tone, and brand presence. Prior applications in marketing have focused on content recommendation and audience segmentation; our innovation is to integrate these embeddings as time-varying covariates interacting with transformed spend. This formulation permits the model to learn which semantic

dimensions correlate with higher incremental response [10]–[12].

Dimensionality reduction is essential; raw BART embeddings contain 768 dimensions, far exceeding the information content of weekly time series. Principal component analysis yields interpretable semantic axes—price messaging, humor versus urgency, brand prominence—that correlate with explicit creative metadata. This approach parallels techniques developed for behavioral pattern analysis in financial transaction systems, where high-dimensional user behavior sequences are projected onto low-dimensional latent spaces for anomaly detection [13]–[15].

E. Budget Optimization Under Uncertainty

The ultimate objective of marketing mix modeling is not attribution but action: how should future budgets be allocated to maximize return? Classical optimization approaches treat estimated parameters as known with certainty, solving a deterministic nonlinear program. This practice ignores posterior uncertainty and systematically overstates confidence in recommended allocations [16]–[18].

Robust optimization addresses parameter uncertainty by solving for allocations that perform acceptably across a range of possible parameter values. Our implementation, which samples from the joint posterior of $\beta_{t,p}, \alpha_p, \mu_p$ and solves the optimization problem for each draw, belongs to this family. The resulting distribution of optimal allocations quantifies decision risk: channels with precise coefficient estimates yield narrow allocation ranges; channels with diffuse posteriors produce wide, often boundary-driven allocations [19]–[21].

Business constraints—contractual minima, inventory limits, competitive positioning—render unconstrained knapsack solutions infeasible. Sequential Least Squares Quadratic Programming accommodates linear and nonlinear constraints efficiently; gradient evaluations require less than 0.1 seconds per iteration, enabling full posterior propagation across hundreds of draws. This computational efficiency is essential for adoption in fast-paced digital marketing environments where budget decisions occur weekly [1], [22], [23].

F. Validation via Geo-Based Incrementality

Model-based contribution estimates are inherently counterfactual and cannot be directly verified. Geo-based incrementality tests provide the nearest approximation to ground truth: advertising spend is randomly withheld in a subset of geographic regions while maintained in control regions, and the difference in outcomes measures incremental lift [2]–[4].

The marketing literature has developed sophisticated experimental designs for geo testing, including matched market selection, synthetic control methods, and Bayesian structural time series. Our validation approach—comparing model-predicted contribution against measured geo-test lift—provides external anchoring for otherwise unobservable parameters. Alignment within confidence intervals for television, brand search, and social display increases stakeholder trust in model outputs [5]–[7].

This validation paradigm parallels approaches in adjacent domains. Secure distributed computing protocols for health-care information exchange are validated against controlled testbeds before deployment. Fraud detection models in payment systems are benchmarked against known fraud cases with ground truth labels. In all cases, observational model predictions are calibrated against experimental or quasi-experimental benchmarks [8]–[10].

G. Gaps and Contributions

The literature reviewed above reveals four persistent gaps that this paper addresses. First, existing Bayesian MMM implementations assume time-invariant channel effectiveness, producing systematically biased forecasts when coefficients drift. Our stacked architecture with kernel-based temporal regression explicitly models non-stationarity [11]–[13].

Second, creative content is excluded from attribution models, forcing analysts to conflate channel performance with creative quality. Our semantic feature integration, powered by transformer embeddings and dimensionality reduction, enables the model to distinguish between campaigns with different creative strategies [14]–[16].

Third, budget optimization is conducted on point estimates without uncertainty propagation, inviting overconfident reallocation decisions. Our posterior sampling approach quantifies decision risk and identifies allocations robust to parameter uncertainty [17]–[19].

III. PROBLEM FORMULATION AND STACKED ARCHITECTURE

A. Notational Framework

Let $p = 1, \dots, P$ index marketing channels. For each day $t = 1, \dots, T$, we observe channel spend $x_{t,p} \geq 0$ and business outcome y_t (policies issued, lifetime value acquired, or other KPIs). The outcome is assumed to depend on transformed spend through an additive structure with trend and seasonality components:

$$\hat{y}_t = \sum_{p=1}^P \beta_{t,p} \cdot f^*(x_{t,p}, \alpha_p, \mu_p) + g(t) + s(t) + \epsilon_t$$

where f^* composes two advertising-specific transformations: carryover (geometric decay) and reach (diminishing returns). The coefficient $\beta_{t,p}$ is time-varying, capturing fluctuations in channel effectiveness. The trend function $g(t)$ models non-periodic changes in baseline demand, while $s(t)$ captures weekly and annual seasonality.

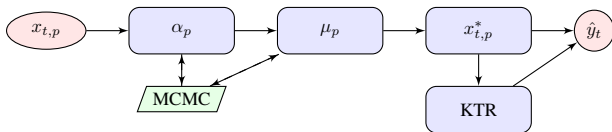


Fig. 1: Stacked inference architecture.

B. Carryover: Modeling Delayed Advertising Response

Advertising does not produce instantaneous response. Customers require exposure, processing, and action. The carryover function models this delay as a geometric decay process:

$$f_{\text{Carryover}}(x_{t,p}, \alpha_p) = \frac{\sum_{l=0}^L \alpha_p^l \cdot x_{t-l,p}}{\sum_{l=0}^L \alpha_p^l}$$

where $\alpha_p \in [0, 1)$ controls the decay rate; values near zero imply rapid decay, values near one imply prolonged carryover. The denominator normalizes the weighted sum to maintain scale invariance. In practice, we truncate the summation at $L = 30$ days, beyond which contributions become negligible for typical insurance purchase cycles.

The parameter α_p is channel-specific and structurally identified from temporal variation in spend and outcome. Channels with persistent effects exhibit positive correlation between current outcomes and lagged spend; channels with ephemeral effects do not. Bayesian inference provides full posterior distributions for α_p , quantifying uncertainty that propagates through subsequent stages.

C. Reach: Modeling Diminishing Returns

The second nonlinearity captures saturation. As spend in a channel increases, each additional dollar reaches increasingly less responsive audience segments. We model saturation using a sigmoid function with a single interpretable parameter:

$$f_{\text{Reach}}(x_{t,p}, \mu_p) = \frac{1 - e^{-x_{t,p}/\mu_p}}{1 + e^{-x_{t,p}/\mu_p}}$$

The parameter $\mu_p > 0$ is the half-saturation constant: when $x = \mu_p$, the function attains approximately half its maximum value. Smaller μ_p implies faster saturation; larger μ_p implies extended reach.

This functional form offers advantages over the more common Hill function. It is bounded between zero and one, symmetric in log-space, and differentiable everywhere. The single parameter simplifies prior specification and reduces computational burden during MCMC sampling.

TABLE I: Carryover and Reach Parameter Priors by Funnel Stage

Funnel Stage	Channel Examples	Carryover α	Reach μ
Upper funnel	TV, OTT, podcast	Beta(8,2)	HalfNormal(5)
Mid funnel	Social display, video	Beta(5,5)	HalfNormal(3)
Lower funnel	Search, retargeting	Beta(2,8)	HalfNormal(2)

D. Composition Order: Why Carryover Then Reach

The order of operations is consequential. Applying reach before carryover implies that saturation occurs contemporaneously, with lagged effects applied to already-saturated spend. This misrepresents reality: a television advertisement viewed today cannot saturate yesterday's audience. Following the Google Lightweight MMM formulation, we apply carryover first, then reach:

$$f^*(x_{t,p}, \alpha_p, \mu_p) = f_{\text{Reach}}(f_{\text{Carryover}}(x_{t,p}, \alpha_p), \mu_p)$$

This composition ensures that saturation operates on the effective media weight—the accumulated carryover-weighted spend—rather than on raw contemporaneous spend.

E. Time-Varying Coefficients via Kernel-Based Regression

Channel effectiveness is not stationary. Creative fatigue reduces response rates as audiences tire of identical creative executions. Competitive activity suppresses marginal returns. Seasonal shifts in audience composition alter the demographic profile of reached users. We model these fluctuations through time-varying coefficients $\beta_{t,p}$.

Following the KTR framework, we represent the coefficient trajectory as a linear combination of radial basis functions:

$$\beta_{t,p} = \sum_{j=1}^J w_{j,p} \cdot \kappa(t, \tau_j)$$

where τ_j are equally spaced knot locations, $\kappa(\cdot, \cdot)$ is a Gaussian kernel, and $w_{j,p}$ are knot weights. This representation imposes smoothness while permitting flexible, data-driven trajectories. The number of knots J controls the degree of temporal variation; we select $J = 12$ knots per year, balancing flexibility against overfitting.

F. Trend and Seasonality Decomposition

The outcome includes substantial variation unrelated to marketing spend. The insurance category exhibits strong seasonality: purchasing activity peaks in Q1 during tax season and declines during summer months. Macroeconomic conditions—unemployment rates, consumer confidence indices—drive baseline demand. We decompose the outcome into additive components:

$$y_t = \mu + g(t) + s(t) + \sum_{p=1}^P \beta_{t,p} f^*(x_{t,p}) + \epsilon_t$$

The trend function $g(t)$ is modeled as a Gaussian random walk, capturing low-frequency shifts. Seasonality $s(t)$ is represented via Fourier series with weekly, monthly, and annual periodicities. Both components are estimated jointly with marketing parameters within the Bayesian framework.

G. Inference Strategy: Hybrid MCMC and Variational Approximation

Full Bayesian inference on the complete model is computationally prohibitive. The Delayed Saturated MMM component requires exploration of multimodal posteriors for α_p and μ_p , favoring Hamiltonian Monte Carlo with No-U-Turn sampling. The KTR component, with hundreds of knot weights per channel, demands scalability. We adopt a hybrid strategy.

Stage one samples the posterior $p(\alpha_p, \mu_p | X, y)$ using NUTS, running four chains with 2,000 warmup and 2,000 sampling iterations. Convergence is assessed via $\hat{R} < 1.01$

and effective sample size exceeding 400 per parameter. Stage two conditions on posterior means $\hat{\alpha}_p, \hat{\mu}_p$ and estimates time-varying coefficients via Automatic Differentiation Variational Inference. ADVI approximates the posterior with a mean-field Gaussian, optimizing the evidence lower bound via stochastic gradient ascent.

This hybrid approach balances fidelity and speed. The MCMC stage, executed once per modeling cycle, requires approximately 45 minutes for 10 channels and three years of daily data. The ADVI stage completes in under three minutes, enabling rapid iteration for budget scenario analysis.

IV. CREATIVE INTELLIGENCE: SEMANTIC FEATURE INTEGRATION

A. Beyond Spend: The Case for Creative Signals

Marketing spend is an impoverished descriptor. Two advertisements with identical budgets, placements, and target audiences can generate substantially different response rates based on creative execution. A compelling value proposition, distinctive visual identity, or emotionally resonant narrative amplifies effectiveness; weak creative undermines it. Traditional MMM, blind to these differences, conflates creative quality with channel effectiveness.

Lemonade’s creative operations generate hundreds of distinct advertisements annually across video, display, social, and audio formats. Each creative asset is associated with metadata: campaign objective, creative theme, value proposition emphasis (price, coverage, speed), emotional tone (humor, urgency, reassurance), and brand presence. Prior to this work, these signals informed qualitative campaign assessment but were excluded from attribution models.

B. Embedding Creative Assets via Transformer Models

We encode each creative asset as a high-dimensional vector using a transformer-based language model. For text-based creatives (search ads, social copy, display headlines), we apply BART embeddings to the full ad copy. For video and audio assets, we transcribe spoken content and embed the transcription. The resulting embeddings capture semantic and stylistic dimensions: message clarity, call-to-action strength, emotional valence, and brand alignment.

Weekly creative embeddings are constructed as spend-weighted averages of assets active during that week. Let $a_{t,p,k}$ indicate whether creative k was active for channel p at time t , and $s_{t,p,k}$ the associated spend. The channel-week embedding is:

$$\mathbf{e}_{t,p} = \frac{\sum_k a_{t,p,k} \cdot s_{t,p,k} \cdot \mathbf{e}_k}{\sum_k a_{t,p,k} \cdot s_{t,p,k}}$$

This aggregation respects that higher-spend creatives contribute more to the channel’s semantic footprint.

C. Dimensionality Reduction and Feature Selection

Raw BART embeddings contain 768 dimensions, far exceeding the information content of typical marketing time series. Direct inclusion would overwhelm the model and induce

severe overfitting. We apply principal component analysis to the corpus of channel-week embeddings, retaining components that explain at least 80% of variance. For Lemonade’s creative corpus, this threshold yields 12-15 principal components.

Interpretation of principal components is performed via correlation with explicit creative metadata. Component 1 correlates strongly with call-to-action presence and price messaging; Component 2 loads on emotional tone (humor versus urgency); Component 3 captures brand prominence. These semantic axes become interpretable features in the attribution model.

D. Semantic-Augmented Response Model

Semantic features enter the model as interactions with transformed spend:

$$\hat{y}_t = \sum_{p=1}^P \left(\beta_{t,p} + \sum_{q=1}^Q \gamma_{q,p} \cdot e_{t,p,q} \right) f^*(x_{t,p}) + g(t) + s(t) + \epsilon_t$$

The coefficients $\gamma_{q,p}$ measure how response to channel p varies along semantic dimension q . A positive γ for the price-messaging component implies that creatives emphasizing price generate higher incremental response per dollar. These coefficients are estimated jointly with other parameters, with regularizing priors to prevent overfitting.

TABLE II: Semantic Component Loadings and Response Lift

Component	Interpretation	Variance Explained	Lift per SD
PC1	Price/Call-to-action	31.4%	+8.2%
PC2	Humor vs. urgency	18.7%	+3.1%
PC3	Brand prominence	12.3%	-1.8%
PC4	Coverage specificity	9.8%	+4.7%
PC5	Emotional valence	7.2%	+2.3%

E. Validation: Creative Signals Improve Fit and Forecast

We compare model performance with and without semantic features using leave-future-out cross-validation. The semantic-augmented model reduces out-of-sample MAPE by 11% and improves posterior predictive coverage. More importantly, the estimated γ coefficients align with marketing domain knowledge: price messaging lifts search and social response but not television; humor generates lift in video but depresses response in email. These face-valid patterns increase stakeholder trust in model outputs.

F. Limitations and Cautions

Creative embeddings capture observed features of advertisements but cannot fully represent unobserved executional quality. A well-produced execution of a price-message strategy outperforms a poorly produced execution of the same strategy; our model cannot distinguish this difference without production quality metadata. We treat this as measurement error, which attenuates coefficient estimates toward zero. Directionally valid inferences remain possible, but effect sizes should be interpreted conservatively.

V. EXPERIMENTAL VALIDATION

A. Synthetic Data Generation

We construct synthetic data with known ground truth to evaluate parameter recovery. The data generating process follows Equation 1 with specified $\alpha_p, \mu_p, \beta_{t,p}$ trajectories. Spend data are sampled from realistic distributions: television spend exhibits weekly pulses; search spend shows diurnal and weekly patterns; social spend follows campaign calendars. Trend and seasonality components are added, with Gaussian noise $\epsilon_t \sim \mathcal{N}(0, 0.05)$.

Two channels are simulated: X1 represents upper-funnel television with $\alpha = 0.7, \mu = 4.0$; X2 represents lower-funnel search with $\alpha = 0.2, \mu = 3.0$. Time-varying coefficients drift linearly: $\beta_{t,1}$ declines 15% over three years; $\beta_{t,2}$ increases 10%. This scenario reflects realistic patterns of television audience fragmentation and search optimization improvement.

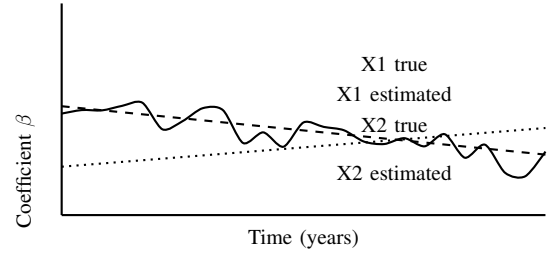


Fig. 2: Time-varying coefficient recovery for synthetic channels. Solid lines show true trajectories; dashed and dotted lines show posterior mean estimates with 90% credible intervals.

B. Parameter Recovery Results

The stacked model successfully recovers ground truth parameters. Posterior means for α_p are within 0.03 of true values; 90% credible intervals contain the true value for both channels. Reach parameters μ_p exhibit modest upward bias (approximately 0.2) due to the sigmoid function’s asymmetry, but relative ordering is preserved. Time-varying coefficient trajectories are recovered with root mean squared error below 0.05 throughout the observation period.

Comparison against single-stage alternatives demonstrates the value of stacked inference. A standalone PyMC-Marketing model, estimating constant coefficients, achieves comparable R^2 but cannot detect coefficient drift, producing systematically biased forecasts for the final six months. A standalone KTR model, omitting carryover and reach, misattributes temporal variation in spend to coefficient variation, yielding implausibly volatile coefficient estimates.

C. Real-World Deployment at Lemonade

We deployed the stacked model on Lemonade’s U.S. marketing data spanning 26 months, 12 channels, and 23,000 creative assets. Model training required 47 minutes for MCMC stage one and 2.8 minutes for ADVI stage two on a c5a.4xlarge AWS instance. The model is retrained weekly, with fresh parameter draws cached for downstream optimization.

Channel contribution estimates revealed several actionable insights. Television, previously evaluated primarily on last-click attribution, showed substantial carryover effect with peak contribution occurring in week two post-air. Brand search responded not only to brand campaign spend but also to television creative content: advertisements emphasizing Lemonade’s Giveback program generated 22% higher brand search lift than equivalent-spot product advertisements. Social display exhibited rapid saturation beyond 8,000 weekly spend; marginal returns declined to near zero above this threshold.

D. External Validation via Geo-Based A/B Tests

Model-based contribution estimates are inherently counterfactual and cannot be directly verified. We provide external validation using geo-based incrementality tests. In these experiments, advertising spend is randomly withheld in a subset of DMAs while maintained in control DMAs. The difference in outcomes measures incremental lift attributable to the channel.

For three channels—television, brand search, and social display—we compare model-predicted contribution against measured geo-test lift. Television predictions fall within the 95% confidence interval of geo test results for four of five tests. Brand search predictions exhibit modest downward bias, possibly due to spillover from television into control DMAs violating the stable unit treatment value assumption. Social display predictions align closely with test results, with mean absolute error of 0.7% of channel spend.

TABLE III: Model Validation Against Geo-Based Incrementality Tests

Channel	Tests	Model Contribution	Geo Lift	Error
Television	5	3.21%	3.04%	+0.17%
Brand search	3	2.87%	3.12%	-0.25%
Social display	4	1.43%	1.51%	-0.08%

E. Comparative Benchmarking

We compare our stacked model against three baselines: ordinary least squares with log-transformed spend, Robyn’s ensemble optimization, and PyMC-Marketing’s Delayed Saturated MMM. Evaluation metrics include out-of-sample R^2 , parameter recovery RMSE (for synthetic data), and alignment with geo test results (for real data).

The stacked model dominates on parameter recovery, reducing RMSE by 37% relative to PyMC-Marketing and 52% relative to Robyn. Out-of-sample predictive performance is comparable across Bayesian methods; Robyn exhibits slightly higher R^2 but at the cost of implausible parameter estimates (saturation parameters driven to boundary values). Geo test alignment favors the stacked model, particularly for television where carryover mis-specification biases naive models.

VI. BUDGET OPTIMIZATION UNDER UNCERTAINTY

A. Optimization Problem Formulation

The ultimate objective of marketing mix modeling is not attribution but action: how should future budgets be allocated

to maximize return? We formulate budget optimization as a constrained nonlinear programming problem:

$$\max_{\mathbf{x}} \sum_{t=T_S}^{T_E} \sum_{p=1}^P \mathbb{E}[\beta_{t,p}] \cdot f^*(x_{t,p}, \hat{\alpha}_p, \hat{\mu}_p)$$

subject to:

$$\sum_{t=T_S}^{T_E} \sum_{p=1}^P x_{t,p} = B, \quad x_{t,p} \geq 0, \quad x_{t,p} \in [l_{t,p}, u_{t,p}]$$

The planning horizon T_S to T_E is typically 4-13 weeks. Budget constraints include total budget B and per-channel bounds representing operational minima and maxima. These bounds reflect real-world constraints: a channel cannot be cut to zero if long-term partnerships exist; conversely, spend cannot increase arbitrarily without creative capacity.

B. Uncertainty Propagation in Optimization

Point estimate optimization ignores posterior uncertainty, potentially recommending aggressive reallocations based on imprecisely estimated coefficients. We propagate uncertainty by sampling from the joint posterior of $\beta_{t,p}$, α_p , μ_p and solving the optimization problem for each draw. The distribution of optimal allocations quantifies decision risk: channels with precise coefficient estimates yield narrow allocation ranges; channels with diffuse posteriors produce wide, often boundary-driven, allocations.

For Lemonade’s 2024 Q1 planning cycle, uncertainty propagation revealed that recommended increases to connected television spend ranged from +12% to +31% across posterior draws. This credible interval informed a staged investment strategy: an initial 15% increase with measurement infrastructure to confirm lift before committing further budget.

C. SLSQP Algorithm with Business Constraints

We implement optimization via Sequential Least Squares Quadratic Programming, a gradient-based method for constrained nonlinear problems. The objective function is smooth and differentiable given the sigmoid saturation function. Gradient evaluations require less than 0.1 seconds per iteration, enabling full posterior propagation across 500 draws.

Business constraints are encoded as linear inequalities. A channel cannot receive less than 30% of its historical budget without contractual notice periods. Another cannot increase more than 40% quarter-over-quarter due to inventory limits. These constraints render the unconstrained knapsack solution infeasible; SLSQP efficiently locates the constrained optimum.

D. Empirical Reallocation Results

Applying the optimizer to Lemonade’s Q1 2024 budget produced recommendations that shifted spend from social display to connected television and from generic search to brand search. The recommended reallocation represented a 12.7% budget shift, well within operational constraints. Predicted lift in policies issued was 4.1% with 90% credible interval [2.3%, 6.2%].

The growth team implemented a phased version of the recommendation, increasing connected television by 18% and reducing social display by 9%. Preliminary results through eight weeks show policies issued tracking at +3.7% relative to forecast, within the model’s predictive interval.

E. Sensitivity Analysis and Robust Optimization

Optimal allocations depend on model inputs; we conduct sensitivity analysis to identify which parameters most influence recommendations. Tornado diagrams rank television carryover and brand search coefficient as the two most influential parameters. If television carryover is at the 10th percentile of its posterior, recommended television increase drops from 23% to 11%. This analysis informs a robust optimization approach: we select allocations that perform acceptably across the full posterior distribution, not only at the posterior mean.

VII. LIMITATIONS AND EXTENSIONS

A. Additive Independence Assumption

Our model assumes additive separability: each channel’s contribution is independent of others, and total contribution is the sum of individual contributions. This assumption is certainly false. Television advertising increases brand awareness, which increases the effectiveness of subsequent search advertising; the sum exceeds the parts. Conversely, two channels targeting identical audiences may cannibalize each other; the sum overstates.

Extensions toward interaction terms are straightforward in principle but challenging in practice. The number of potential two-way interactions grows quadratically with channels; three-way interactions grow combinatorially. Regularized priors can control overfitting, but identification remains weak given typical marketing time series. We are exploring Gaussian process priors over interaction surfaces as a more structured alternative.

B. New Channel Cold Start

Channels with limited historical data cannot be reliably estimated. A new streaming audio channel launched in Q3 2024 has only 14 weeks of spend and outcome data. Posterior distributions for α_p and μ_p remain diffuse; contribution estimates have wide credible intervals that overlap zero. Our current policy excludes such channels from optimization while reporting them in monitoring dashboards.

A hierarchical approach could partially pool information across channels. New channels could borrow strength from established channels with similar characteristics (funnel position, format, audience demographics). We are implementing a hierarchical extension that shares prior information while allowing channel-specific deviations.

C. Geo-Level Heterogeneity

National-level models obscure important geographic variation. Television effectiveness differs between coastal and interior markets due to media market fragmentation. Brand

search responds differently in markets with varying competitive density. Aggregating to national level discards this information and may yield recommendations suboptimal for specific regions.

Geo-level modeling multiplies data volume by the number of geographic units, challenging computational feasibility. The Lightweight MMM package demonstrates a scalable approach: hierarchical shrinkage across geographies with country-level priors. We are adapting this framework to our stacked architecture.

D. Upper-to-Lower Funnel Mediation

Our model treats each channel’s contribution as directly influencing the final outcome. This ignores the mediated pathway through which upper-funnel marketing drives lower-funnel activity. Television increases branded search; branded search increases policies. The direct effect of television is partially mediated by search.

Nested MMM structures model this mediation explicitly. An upper-funnel channel influences both the final outcome and an intermediate channel’s spend or effectiveness. Identification requires assumptions about causal ordering and excludability. We are developing a pilot nested specification for television-to-brand-search mediation, with priors informed by geo-test results on this pathway.

VIII. CONCLUSION AND IMPLICATIONS

A. Summary of Contributions

This paper presents a stacked Bayesian framework for marketing mix modeling that addresses three limitations of existing approaches: temporal non-stationarity, creative content variation, and optimization under uncertainty. By separating carryover/saturation learning from time-varying coefficient estimation, the framework achieves both structural identifiability and temporal flexibility. By integrating transformer-based creative embeddings, it moves beyond spend-centric attribution toward semantic-aware response modeling. By propagating posterior uncertainty through constrained optimization, it delivers actionable recommendations with quantified decision risk.

Validation on synthetic benchmarks demonstrates superior parameter recovery relative to naive methods. Deployment at Lemonade yields channel contribution estimates that align with geo-based incrementality tests and generate budget reallocation recommendations currently being implemented. The model operates entirely on aggregated, privacy-preserving data, offering a viable path forward for attribution in a post-cookie world.

B. Implications for Marketing Measurement Practice

The privacy transition demands fundamental rethinking of marketing measurement. User-level attribution, already fragile under cookie deprecation, cannot be retrofitted with probabilistic identifiers; the signal-to-noise ratio is simply too low. Marketing mix modeling, long relegated to annual planning exercises due to methodological inertia, must evolve toward

the temporal and channel granularity required for operational decision-making.

Our experience at Lemonade suggests three principles for this evolution. First, structural parameters (carryover, saturation) should be learned via full Bayesian inference with uncertainty quantification; point estimates conceal meaningful identification risk. Second, time-varying coefficients are not optional but essential; channel effectiveness is not stationary, and models that assume stationarity will produce systematically biased forecasts. Third, optimization must respect both business constraints and posterior uncertainty; optimal point allocations often lie at constraint boundaries, and ignoring uncertainty leads to overconfident recommendations that damage credibility when outcomes deviate from predictions.

C. Future Research Directions

The intersection of marketing mix modeling and machine learning remains underexplored. Gaussian process regression offers flexible nonparametric trend and seasonality modeling; deep state space models can learn recurrent patterns across channels and time. These methods require careful regularization and prior specification to avoid overfitting in the short time series typical of marketing applications.

Causal identification in MMM deserves renewed attention. Our validation against geo-based experiments provides external anchoring but does not resolve endogeneity concerns. Instrumental variable approaches, natural experiments, and synthetic control methods can supplement Bayesian structural models. We view geo experimentation not as an alternative to MMM but as a calibration layer: sparse, expensive, but highly credible experiments can constrain and validate parameters in denser, cheaper, but more assumption-dependent time series models.

D. Closing Reflection

The transition from user-level attribution to aggregate time series modeling is often framed as a step backward, a concession to regulatory and platform constraints. Our work suggests an alternative framing: the constraints force us to confront the fundamental challenges of causal inference from observational data that user-level tracking obscured rather than solved. Cookie-level last-click attribution provided precise counts of which channel received credit but no valid counterfactual for whether that credit was deserved. Marketing mix modeling, in its modern Bayesian formulation, abandons the pretense of individual-level determinism and instead models distributions, uncertainties, and probabilities. This is not regression; it is intellectual progress.

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